3	Two spheres with different masses are initially moving towards each other with the same speed. The two spheres collide elastically. Which of the following statements is **true**?	
	A	The two spheres can be at rest at the same time because they were initially moving in opposite directions.
	B	The two spheres can be at rest at the same time because momentum is always conserved in an elastic collision.
	C	The two spheres cannot be at rest at the same time at any point in time because the total momentum of the system is not zero.
	D	The two spheres cannot be at rest at the same time at any point in time because energy is always conserved in an elastic collision.
	Solution: C Since the two spheres do not have the same momentum (since their momentum are unequal in magnitudes), the total momentum of the system is not zero, and hence they cannot be at rest at the same time.	

4	A rower is sitting on a boat in the middle of a calm lake.